A UNIVERSAL PHANTOM SUBCATEGORY ON THE BLOW-UP OF $\mathbb{P}^2_{\mathbb{C}}$ AT TEN GENERAL POINTS

ARGUS OPEN RESEARCH PROJECT

Abstract. We document Krah's 2024 theorem proving that on the rational surface X obtained by blowing up $\mathbb{P}^2_{\mathbb{C}}$ at 10 general points, there exists an exceptional collection of 13 line bundles of maximal length whose orthogonal complement in $D^b(X)$ is a nonzero universal phantom category.

1. Main Refutation

Theorem 1.1 (Krah, 2024)**.** *Let X be the blow-up of $\mathbb{P}^2_{\mathbb{C}}$ at 10 general points. Although $D^b(X)$ admits full exceptional collections of length $13 = \mathrm{rk}\, K_0(X)$, there exists an exceptional collection $\mathcal{E} = (L_1, \ldots, L_{13})$ of line bundles on X that is not full: its orthogonal complement $\mathcal{A} = \langle \mathcal{E} \rangle^{\perp} \subset D^b(X)$ is a nonzero universal phantom subcategory (having vanishing Hochschild homology and $K_0(\mathcal{A} \boxtimes D^b(Y)) = 0$ for every smooth projective variety Y).*

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